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Supplementary Information: experimental experience and scientific understanding

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Retained numerical evidence. Research cohorts and all failure dispositions remain separate.

Supplementary Information

Study organization

The broad study (300 campaigns), earlier selected notebook studies (24 sessions), and repeated process cohort (90 sessions) retain separate denominators and execution provenance. Reference batches, recommendation retests and fixed-recipe controls are not researcher sessions. The process cohort retains 84 complete chains, 1,079/1,080 source batches, 298/315 assessment stages and 86 valid forecast stages. Notebook reading covers all 284 revisions, but qualitative interpretation is by one analyst and selected cases do not establish mechanism prevalence. The separate main-text PDF and this Supplementary Information form one reading package; the main-text file alone does not contain the full tables.

Recovery definitions and the affected readouts

The full-process observation response used in source assays, recommendation retests and prediction references is seed-excluded solid target divided by the tracked initial reagent amount, clipped to [0,1]. It is distinct from the internal crystallization-stage metric and from the definition in the saved participant prompts.

Quantity or interface	Denominator	Role in the retained study
Internal crystallization module	Target product before separation	Stage-recovery metric, subsequently overridden for full-process observation
Source final assay	Tracked initial reagent amount	Noisy full-process observation provided to the researcher
Recommendation and query reference	Tracked initial reagent amount	Noiseless full-process response captured before final-assay sampling
Forecast evaluator	Same captured initial-charge response	Original errors and interval coverage; scores retained unchanged
Source and forecast prompts	Target product before separation	Participant-facing description mismatched to the observed/scored output

The numerator excludes seed, but matching numerators do not make these denominators equivalent. This mismatch occurs in both recorded process execution versions; it is not a verified historical-to-current normalization change. Recovery scores and joint quality/recovery counts remain reproducible protocol-affected diagnostics. They are not evidence of performance on an unambiguously specified recovery target. Particle/purity/fines feasibility is reported separately: 10/15 Opaque, 9/15 Aligned and 14/15 MisIndexed, compared with retained joint counts 9/15, 7/15 and 14/15. Purity and fines do not use the disputed denominator, but the research paths were generated under the original instructions. No original prediction, reference or score has been changed.

Complete equilibrium process outcomes

Each row is one independent model session within a fixed world and information condition. O/A/M denote Opaque/Aligned/MisIndexed. A dash is a missing forecast. All M sessions belong to the extension cohort; O/A sessions belong to the first cohort. MAEs average the three equilibrium response channels. The other-nine group is not an in-distribution control.

World	Arm	Repeat	Batches	Other-nine MAE	Dilute MAE	Dilute coverage
01	O	1	12	0.00650	0.20238	0.44444
01	O	2	12	0.01577	0.09224	1.00000
01	O	3	12	-	-	-
01	A	1	12	0.00642	0.19140	0.00000
01	A	2	12	0.00371	0.04234	0.77778
01	A	3	12	-	-	-
01	M	1	12	0.00590	0.18631	0.11111
01	M	2	11	-	-	-
01	M	3	12	0.00556	0.17253	0.00000
02	O	1	12	0.00537	0.17576	1.00000
02	O	2	12	0.01101	0.01501	0.91111
02	O	3	12	0.01051	0.01451	1.00000
02	A	1	12	0.00521	0.16909	0.00000
02	A	2	12	0.00787	0.16609	0.11111
02	A	3	12	0.00794	0.01350	1.00000
02	M	1	12	0.00571	0.17803	0.00000
02	M	2	12	0.00699	0.18604	0.00000
02	M	3	12	0.00561	0.18310	0.00000
03	O	1	12	0.00907	0.14145	0.11111
03	O	2	12	0.02607	0.01273	0.91111
03	O	3	12	0.00399	0.01158	1.00000
03	A	1	12	0.00514	0.16287	0.00000
03	A	2	12	0.00431	0.15325	0.00000
03	A	3	12	0.00361	0.15504	0.00000
03	M	1	12	0.00363	0.13845	0.60000
03	M	2	12	0.00383	0.15885	0.00000
03	M	3	12	0.00432	0.15848	0.00000
04	O	1	12	0.00496	0.04810	0.80000
04	O	2	12	0.00697	0.14143	0.66667
04	O	3	12	0.00756	0.11121	0.80000
04	A	1	12	0.00465	0.14272	0.00000
04	A	2	12	0.00471	0.14012	0.00000
04	A	3	12	-	-	-
04	M	1	12	0.00498	0.14234	0.00000
04	M	2	12	0.00769	0.14730	0.00000
04	M	3	12	0.00364	0.10860	0.33333
05	O	1	12	0.00377	0.14630	0.71111
05	O	2	12	0.00818	0.08730	1.00000
05	O	3	12	0.00515	0.13473	0.00000
05	A	1	12	0.00424	0.12640	0.64444
05	A	2	12	0.00383	0.13206	0.00000
05	A	3	12	0.00546	0.12764	0.00000
05	M	1	12	0.00821	0.06752	0.88889
05	M	2	12	0.00502	0.11362	0.00000
05	M	3	12	0.00281	0.10662	0.24444

Complete crystallization process outcomes

All rows have twelve source batches and valid forecasts. Success applies the original joint recommendation criteria to the noiseless retest. All M sessions, and all W02/W04 sessions, belong to the extension cohort. Other O/A sessions belong to the first cohort. Recovery MAE and Success retain the implemented initial-charge target under the mismatched prompt, as detailed above; neither is a clean recovery-understanding readout.

World	Arm	Repeat	Success	Purity MAE	Recovery MAE	Fines MAE
01	O	1	No	0.01514	0.07036	0.49144
01	O	2	Yes	0.01494	0.19041	0.37607
01	O	3	Yes	0.01529	0.05894	0.26897
01	A	1	No	0.06045	0.20875	0.39186

World	Arm	Repeat	Success	Purity MAE	Recovery MAE	Fines MAE
01	A	2	No	0.07045	0.08237	0.35256
01	A	3	Yes	0.00811	0.16938	0.41231
01	M	1	Yes	0.04269	0.07619	0.29191
01	M	2	Yes	0.06712	0.14437	0.17320
01	M	3	Yes	0.04295	0.05451	0.41392
02	O	1	No	0.02065	0.05299	0.34607
02	O	2	Yes	0.02067	0.21933	0.29524
02	O	3	Yes	0.01292	0.15573	0.41107
02	A	1	No	0.00615	0.12426	0.24719
02	A	2	No	0.07982	0.22558	0.37703
02	A	3	No	0.00551	0.11615	0.36440
02	M	1	Yes	0.07148	0.10312	0.51203
02	M	2	Yes	0.07898	0.06437	0.26370
02	M	3	Yes	0.01983	0.17259	0.38024
03	O	1	No	0.00939	0.17261	0.39280
03	O	2	Yes	0.02235	0.05261	0.30280
03	O	3	Yes	0.03235	0.12178	0.33864
03	A	1	No	0.02652	0.15761	0.24197
03	A	2	Yes	0.08319	0.07243	0.35403
03	A	3	Yes	0.03569	0.05836	0.32115
03	M	1	No	0.04944	0.06940	0.30115
03	M	2	Yes	0.04319	0.06980	0.27032
03	M	3	Yes	0.06485	0.05317	0.32115
04	O	1	Yes	0.01027	0.04411	0.24413
04	O	2	No	0.01105	0.11431	0.24150
04	O	3	Yes	0.02029	0.15714	0.26580
04	A	1	Yes	0.09446	0.07348	0.21295
04	A	2	Yes	0.04029	0.10416	0.22004
04	A	3	No	0.00994	0.05034	0.29795
04	M	1	Yes	0.11529	0.04371	0.26628
04	M	2	Yes	0.12113	0.04513	0.19711
04	M	3	Yes	0.08529	0.07505	0.26128
05	O	1	No	0.00534	0.09232	0.14739
05	O	2	No	0.00543	0.24727	0.31748
05	O	3	Yes	0.01443	0.05151	0.32656
05	A	1	Yes	0.01246	0.07328	0.28195
05	A	2	Yes	0.02060	0.08932	0.29656
05	A	3	No	0.06445	0.32577	0.25906
05	M	1	Yes	0.02927	0.11660	0.28520
05	M	2	Yes	0.04528	0.10225	0.32187
05	M	3	Yes	0.01778	0.13283	0.24990

Incomplete chains

All six incomplete chains are retained. Missing stages do not imply a forecast of zero. K1 is the scientific account, Q the prediction, K2 the retrospective, and EQS the structured equilibrium supplement. Five chains lose one or more assessment payloads. The extension W01/M repeat 2 terminates after eleven batches on a resource-accounting validation failure; none of its assessment stages is available.

World	Arm	Repeat	Cohort	Missing stages
05	O	2	First	K2, EQS
01	O	3	First	Q, K2, EQS
01	A	3	First	K1, Q, K2, EQS
04	A	3	First	Q, K2, EQS
03	M	1	Extension	EQS
01	M	2	Extension	K1, Q, K2, EQS

Scope of process cases

Main Figures 2-4 and Extended Data Figures 1-2 connect original public records to executed actions. The equilibrium mapping case is world 4, M, repeat 1; the seed-rescue case world 1, M, repeat 3; the volume-window case world 1, M, repeat 2. The purity comparison retains all three world-4/M sessions. The incorrect cumulative-seed account is world 4, A, repeat 3; the rejection-recognizing counterexample is world 2, A, repeat 2. Risk-choice examples are world 2, A, repeat 3 and world 4, A, repeat 2. These labels locate evidence; they do not define participant subpopulations or prevalence denominators.

Record availability across all process sessions

This inventory reports available records, not the presence or absence of a scientific mechanism. All initial notes precede physical operations under the prompted notebook role. Notes counts include the initial version. K1 is the sealed source account, Q the forecast, K2 the retrospective, and EQS the equilibrium supplement. Unavailable stages remain missing. A session not selected below has not been classified as a negative case. World and arm labels have the same meaning as in the outcome tables.

System	World	Arm	Repeat	Notes	Initial before action	Stages
C	01	O	1	5	Yes	K1, Q, K2
C	01	O	2	3	Yes	K1, Q, K2
C	01	O	3	6	Yes	K1, Q, K2
C	01	A	1	6	Yes	K1, Q, K2
C	01	A	2	3	Yes	K1, Q, K2
C	01	A	3	3	Yes	K1, Q, K2
C	01	M	1	4	Yes	K1, Q, K2
C	01	M	2	5	Yes	K1, Q, K2
C	01	M	3	4	Yes	K1, Q, K2
C	02	O	1	2	Yes	K1, Q, K2
C	02	O	2	2	Yes	K1, Q, K2
C	02	O	3	3	Yes	K1, Q, K2
C	02	A	1	2	Yes	K1, Q, K2
C	02	A	2	3	Yes	K1, Q, K2
C	02	A	3	4	Yes	K1, Q, K2
C	02	M	1	3	Yes	K1, Q, K2
C	02	M	2	5	Yes	K1, Q, K2
C	02	M	3	2	Yes	K1, Q, K2
C	03	O	1	4	Yes	K1, Q, K2
C	03	O	2	3	Yes	K1, Q, K2
C	03	O	3	3	Yes	K1, Q, K2
C	03	A	1	5	Yes	K1, Q, K2
C	03	A	2	3	Yes	K1, Q, K2
C	03	A	3	4	Yes	K1, Q, K2
C	03	M	1	2	Yes	K1, Q, K2
C	03	M	2	3	Yes	K1, Q, K2
C	03	M	3	3	Yes	K1, Q, K2
C	04	O	1	4	Yes	K1, Q, K2
C	04	O	2	3	Yes	K1, Q, K2
C	04	O	3	3	Yes	K1, Q, K2
C	04	A	1	4	Yes	K1, Q, K2
C	04	A	2	2	Yes	K1, Q, K2
C	04	A	3	6	Yes	K1, Q, K2
C	04	M	1	3	Yes	K1, Q, K2
C	04	M	2	4	Yes	K1, Q, K2
C	04	M	3	4	Yes	K1, Q, K2
C	05	O	1	4	Yes	K1, Q, K2
C	05	O	2	3	Yes	K1, Q, K2
C	05	O	3	3	Yes	K1, Q, K2
C	05	A	1	5	Yes	K1, Q, K2
C	05	A	2	3	Yes	K1, Q, K2
C	05	A	3	3	Yes	K1, Q, K2

System	World	Arm	Repeat	Notes	Initial before action	Stages
C	05	M	1	4	Yes	K1, Q, K2
C	05	M	2	5	Yes	K1, Q, K2
C	05	M	3	3	Yes	K1, Q, K2
EQ	01	O	1	2	Yes	K1, Q, K2, EQS
EQ	01	O	2	3	Yes	K1, Q, K2, EQS
EQ	01	O	3	2	Yes	K1
EQ	01	A	1	2	Yes	K1, Q, K2, EQS
EQ	01	A	2	2	Yes	K1, Q, K2, EQS
EQ	01	A	3	2	Yes	None
EQ	01	M	1	3	Yes	K1, Q, K2, EQS
EQ	01	M	2	2	Yes	None
EQ	01	M	3	4	Yes	K1, Q, K2, EQS
EQ	02	O	1	3	Yes	K1, Q, K2, EQS
EQ	02	O	2	3	Yes	K1, Q, K2, EQS
EQ	02	O	3	3	Yes	K1, Q, K2, EQS
EQ	02	A	1	2	Yes	K1, Q, K2, EQS
EQ	02	A	2	3	Yes	K1, Q, K2, EQS
EQ	02	A	3	3	Yes	K1, Q, K2, EQS
EQ	02	M	1	3	Yes	K1, Q, K2, EQS
EQ	02	M	2	3	Yes	K1, Q, K2, EQS
EQ	02	M	3	3	Yes	K1, Q, K2, EQS
EQ	03	O	1	2	Yes	K1, Q, K2, EQS
EQ	03	O	2	3	Yes	K1, Q, K2, EQS
EQ	03	O	3	2	Yes	K1, Q, K2, EQS
EQ	03	A	1	3	Yes	K1, Q, K2, EQS
EQ	03	A	2	2	Yes	K1, Q, K2, EQS
EQ	03	A	3	3	Yes	K1, Q, K2, EQS
EQ	03	M	1	4	Yes	K1, Q, K2
EQ	03	M	2	3	Yes	K1, Q, K2, EQS
EQ	03	M	3	3	Yes	K1, Q, K2, EQS
EQ	04	O	1	3	Yes	K1, Q, K2, EQS
EQ	04	O	2	2	Yes	K1, Q, K2, EQS
EQ	04	O	3	3	Yes	K1, Q, K2, EQS
EQ	04	A	1	3	Yes	K1, Q, K2, EQS
EQ	04	A	2	2	Yes	K1, Q, K2, EQS
EQ	04	A	3	2	Yes	K1
EQ	04	M	1	4	Yes	K1, Q, K2, EQS
EQ	04	M	2	3	Yes	K1, Q, K2, EQS
EQ	04	M	3	3	Yes	K1, Q, K2, EQS
EQ	05	O	1	2	Yes	K1, Q, K2, EQS
EQ	05	O	2	3	Yes	K1, Q
EQ	05	O	3	2	Yes	K1, Q, K2, EQS
EQ	05	A	1	3	Yes	K1, Q, K2, EQS
EQ	05	A	2	3	Yes	K1, Q, K2, EQS
EQ	05	A	3	3	Yes	K1, Q, K2, EQS
EQ	05	M	1	3	Yes	K1, Q, K2, EQS
EQ	05	M	2	4	Yes	K1, Q, K2, EQS
EQ	05	M	3	2	Yes	K1, Q, K2, EQS

Structured focal relationships

Nine selected sessions are organized below by the relationship under examination. Cases 4-6 retain all three sessions in one crystallization world/information cell. Other cases are selected trajectories or counterexamples, not a random sample. Each record separates prior expectation, successful execution, explicit revision, evidential scope and later use. A pre-test claim not established by the focal analysis is labelled accordingly; it is not scored as absent. Quotations are checked against their specified note version or sealed stage. Interpretation remains by one analyst, without independent reliability assessment. All figures remain illustrative previews; these records and the outcome tables carry the retained evidence.

Process case 1: One local test, two evidential roles

EQ, world 4, MisIndexed, session 1. Extension cohort.

Prior expectation. Revision 3, before batches 11-12, predicts a substantial response if the alternative component is the active coordinate.

Executed test or observations. Batch 11 tests the component alone; batch 12 adds a large component spike to a small reagent charge. Both committed tests remain near the observed response plateau.

Recorded revision or characterization. Revision 4 withdraws support for the tested mapping while retaining buffering, saturation and unidentified-coordinate explanations.

What the evidence supports. The executed perturbations oppose the specified mapping prediction. They do not determine the low-loading limit or distinguish a persistent plateau from a dilute transition.

Subsequent use. The extreme-dilution forecast explicitly invokes the component-only result: dissociation 0.092, interval 0.043-0.155, versus reference mean 0.51730. Retrospective review recognizes unresolved low-loading alternatives without receiving truth.

Extent of support. One fully connected focal trajectory; the 45-session equilibrium distribution contextualizes the outcome but does not establish the frequency of this process.

Recorded wording (note version 3). "A null component response would instead favor an internally buffered/invariant world."

Recorded wording (note version 4). "component-as-active-loading hypothesis rejected in tested form"

Recorded wording (Q). "shrink strongly toward the experimentally observed near-invariant baseline because a component-only batch also retained similar responses."

Process case 2: A prospective seed-rescue expectation is withdrawn

C, world 1, MisIndexed, session 3. Extension cohort.

Prior expectation. Revision 3 proposes repeating the concentrated 0.02 L recipe with 0.05 g seed to test whether particle quality can be rescued.

Executed test or observations. Batches 10 and 12 share the principal preparation and cooling operations; committed seed changes from 0.02 to 0.05 g. Both assays report fines fraction 1.0.

Recorded revision or characterization. Revision 4 explicitly disconfirms this prospective rescue expectation and retains a different recipe as the recommendation.

What the evidence supports. The larger of the two executed doses does not rescue the tested condition. No conclusion about all larger doses or every seeding mechanism follows.

Subsequent use. The recommendation uses an alternative recipe whose noiseless fines fraction is 0.24744 and which satisfies the recorded feasibility criteria. Recovery rankings remain subject to the disclosed definition mismatch.

Extent of support. One prospective dose contrast and recorded revision; no independent repeat of this exact hypothesis test is claimed.

Recorded wording (note version 3). "Repeat the 0.02 L high-loading recipe using 0.05 g seed."

Recorded wording (note version 4). "disconfirming the final prospective rescue hypothesis."

Process case 3: A working recipe region guides a new test

C, world 1, MisIndexed, session 2. Extension cohort.

Prior expectation. After a feasible 35 ml recipe, revision 4 proposes 32.5 ml as the next boundary test.

Executed test or observations. Batches 10-12 use 35, 32.5 and 35 ml, with otherwise matching principal operations. Measured fines fractions are 0.14526, 0.03411 and 0.13587.

Recorded revision or characterization. The final note retains a bounded feasible region and selects the repeated 35 ml recipe. It does not claim that small differences among close recovery values are resolved.

What the evidence supports. The sequence supports feasibility of the tested recipes. Earlier low-volume failures also changed seed or thermal history, so a sharp concentration threshold is not isolated.

Subsequent use. The proposed intermediate-volume test is executed and the final recommendation passes an independent retest, with fines fraction 0.13392. There is no clean causal estimate of a universal volume effect.

Extent of support. One prospective test, one same-recipe repeat and one recommendation retest in a single session; these are not independent researcher samples.

Recorded wording (note version 4). “Next test at 0.0325 L targets the boundary.”

Recorded wording (note version 5). “Its 0.035 L recipe was independently repeated in batch 12 and remained compliant”

Process case 4: Recognized high purity and altered transfer, first session

C, world 4, MisIndexed, session 1. Extension cohort.

Prior expectation. A specific pre-test numerical commitment to invariant purity is not established by this focal comparison; the relevant supported judgment appears in the sealed source account.

Executed test or observations. Twelve source assays have mean purity 99.09%; the selected recipe passes the recorded retest criteria.

Recorded revision or characterization. The source account explicitly recognizes impurity rejection and high crystal purity. This is a documented local characterization, not a quantified pre/post learning gain.

What the evidence supports. High purity is observed in the tested recipes. Transfer still requires judging effects of changed conditions; no universal purity law is inferred.

Subsequent use. Forecast adjustments for solvent, occlusion, concentration and thermal history yield mean purity 86.92%, versus reference 98.45%. Purity MAE is 0.11529 and 0/12 purity intervals cover the reference.

Extent of support. One of all three sessions in the same world/information cell; all three are retained below.

Recorded wording (K1). “filtered crystals generally had purity 0.98–1.00”

Process case 5: Recognized high purity and altered transfer, second session

C, world 4, MisIndexed, session 2. Extension cohort.

Prior expectation. A specific pre-test commitment to unchanged purity is not established by this comparison; the source account explicitly characterizes the observed response.

Executed test or observations. Twelve source assays have mean purity 98.18%; the recommendation passes the recorded retest criteria.

Recorded revision or characterization. The report recognizes high purity and a repeatedly feasible process region. It also contains a separate inaccurate seed-dose account; recognizing purity does not validate every other claim in that report.

What the evidence supports. High purity and recipe feasibility are supported locally. The reported large-dose seed comparison requires separate execution checking.

Subsequent use. Mean purity forecast is 86.33%, versus common reference 98.45%. No claim of forgetting or a shared psychological mechanism is made.

Extent of support. Second of the three same-cell sessions, rather than a second independently sampled world.

Recorded wording (K1). “S2 did give high purity and substantial recovery”

Process case 6: Recognized high purity and altered transfer, third session

C, world 4, MisIndexed, session 3. Extension cohort.

Prior expectation. No specific pre-test invariance commitment is assigned; the comparison concerns established source observations and their subsequent use.

Executed test or observations. Twelve source assays have mean purity 99.09%; the recommendation passes the recorded retest criteria.

Recorded revision or characterization. The source account recognizes that observed S2 purities are generally 0.98-1.00 and acknowledges confounding in solvent comparisons.

What the evidence supports. The observed high-purity region is supported; an independent solvent effect is not identified.

Subsequent use. Mean purity forecast is 89.92%, versus reference 98.45%. This repeats the recognition/forecast contrast, not necessarily the route that produced it.

Extent of support. Third of all three same-cell sessions; no population probability is estimated from three repeats.

Recorded wording (K1). “Low impurity occlusion is consistent with the observed final purities, generally 0.98–1.00 in S2 batches.”

Process case 7: An intended large-dose test becomes negative evidence

C, world 4, Aligned, session 3. Extension cohort.

Prior expectation. Revision 4 proposes approximately 2 g cumulative seed, treating 0.05 g as a per-operation allowance.

Executed test or observations. Batch 10 requests forty additions of 0.05 g; one commits and thirty-nine are rejected. Total committed seed is 0.05 g.

Recorded revision or characterization. Revision 5 states that increasing seed to about 2 g did not reduce fines and excludes seed mass as the limiting control. Later notes and sealed accounts retain the claim.

What the evidence supports. The trajectory provides no physical test of 2 g. A failed request is not negative evidence for the scientific dose hypothesis.

Subsequent use. The claimed exclusion persists in the final account and retrospective review. Its independent influence on subsequent actions has not been tested.

Extent of support. One connected execution/report discrepancy, with a rejection-recognizing counterexample in case 8. Rejection counts across the cohort are not error-prevalence estimates.

Recorded wording (note version 4). “Test about 2 g cumulative seed”

Recorded wording (note version 5). “Seed mass is not the limiting control in this range.”

Process case 8: Rejected additions are correctly recognized

C, world 2, Aligned, session 2. Extension cohort.

Prior expectation. The attempted repeated addition is checked against its actual execution status; a separate prospective high-dose scientific expectation is not assigned by this focal analysis.

Executed test or observations. Additional seed requests in batch 2 are rejected by payload bounds.

Recorded revision or characterization. Revision 2 explicitly records the failed additions and states that one legal seed operation should be used per batch unless the range changes.

What the evidence supports. The operational conclusion concerns permitted dosing, not a physical effect of the rejected larger dose.

Subsequent use. The recorded decision limits further dosing rather than claiming a successful large-dose test. A causal effect of the note itself is not estimated.

Extent of support. A selected counterexample showing that rejection need not become a false experimental claim.

Recorded wording (note version 2). “use one legal seed operation per batch unless the current legal range changes.”

Process case 9: Accurate extrapolation with an unresolved mechanism

EQ, world 2, Opaque, session 2. First cohort.

Prior expectation. The source account retains multiple explanations for an observed plateau; a directly executed extreme-dilution discriminator is absent from the source action record.

Executed test or observations. Twelve source batches characterize a plateau without positive-loading assays at or below 1 mM.

Recorded revision or characterization. An effective soluble-concentration ceiling becomes more specific during forecasting, although it had remained one candidate in the source account.

What the evidence supports. The source observations do not uniquely identify the ceiling mechanism. Correct extrapolation can use theoretical structure without direct sampling.

Subsequent use. The dilute forecast is relatively accurate; retrospective review still recognizes that its intervals underrepresent the competing persistent-plateau explanation.

Extent of support. One positive extrapolation counterexample within the first process block; accuracy does not retrospectively make the mechanism experimentally identified.

Recorded wording (K2). “This is inconsistent with K1’s explicit statement that solubility pinning, buffering, nonideality, and synthetic mapping were observationally equivalent”

All matched quench dispositions

Donor numbers below identify the eighteen first-cohort recommendations in a fixed sorted order. Five worlds are reused for each donor. A dash is a missing final paired effect, with the failed arm retained. Effects are present minus absent and use noiseless fines. They describe the complete pairs, not all planned recipes. Invalid arms are not repaired.

Donor	Source world	Information	Repeat
1	01	Aligned	01
2	01	Opaque	01
3	03	Aligned	01
4	03	Opaque	01
5	05	Aligned	01
6	05	Opaque	01
7	01	Aligned	02
8	01	Opaque	02
9	03	Aligned	02
10	03	Opaque	02
11	05	Aligned	02
12	05	Opaque	02
13	01	Aligned	03
14	01	Opaque	03
15	03	Aligned	03
16	03	Opaque	03
17	05	Aligned	03
18	05	Opaque	03

Donor	World 1	World 2	World 3	World 4	World 5
1	Absent fails	Absent fails	Absent fails	Absent fails	Absent fails
2	0.00000	0.00000	0.00000	0.00000	0.00000
3	-0.07193	0.08121	-0.06641	-0.03122	-0.11081
4	0.30483	0.25054	0.26820	0.29006	0.00016
5	0.36050	0.75196	0.74491	0.70227	0.74283
6	0.00000	0.00000	0.00000	0.00000	0.00000
7	0.00000	-0.00147	0.00000	0.00000	0.00000
8	0.73643	0.67614	0.65346	0.61596	0.62552
9	Present fails	Present fails	Present fails	Present fails	Present fails
10	Present fails	Present fails	Present fails	Present fails	Present fails
11	0.17751	0.52630	0.52558	0.49988	0.57637
12	0.21897	0.14705	0.14699	0.07543	0.06037
13	0.00558	-0.05076	-0.09236	-0.08477	-0.21882
14	0.71771	0.64406	0.68547	0.62873	0.64872
15	0.60226	0.00022	0.61733	0.55273	0.34794
16	0.54132	0.42409	0.56943	0.54616	0.59024
17	0.57332	0.37074	0.54879	0.55370	0.49501
18	0.43274	0.47053	0.50706	0.52815	0.56957

Retained broad-study tables

The following numerical tables retain the earlier cross-system and targeted-model results without pooling them with the process cohort. Their original descriptive groupings, task-specific response scales and sample units remain unchanged. E, P and S as information loci denote entity, parameter and structure information; P as a system denotes purification. EC is electrochemistry, PA partitioning, RX reaction, EQ equilibrium and C crystallization. Tables with example trajectories refer to the earlier study, not to the new process cases. SD is descriptive variation, not a confidence interval.

Retained table 1: A.1 Complete study matrix

System / prior locus	Scientific commission	Budgets	Campaigns	Final-assayed batches
EC / E: electrochemistry	Discovery; score optimization	12, 24	60	1,080 / 1,080
PA / E: phase partitioning	Explain phase allocation	12, 24	30	540 / 540
RX / P and S: reaction and thermal processing	Discovery; safety-constrained optimization	12	60	720 / 720
EQ / E: medium identity	Characterize entity-conditioned responses	12	15	180 / 180
EQ / P: bounded equilibrium	Characterize effective responses	12	15	180 / 180
EQ / S: structural equilibrium	Explain response structure	12	15	180 / 180
C / E: crystallization	Deliver crystals under quality constraints	12, 24	30	539 / 540
P / E: purification	Recover product subject to purity	12	15	178 / 180
Total	**Six system families**		**240**	**3,597 / 3,600**

Table A1. Cross-system study scope (GPT-5.6 Sol, medium). Each campaign has one agent realization. One C campaign exhausts its solvent stock and discards its twelfth vessel, completing eleven final assays. One P campaign uses all twelve vessel starts but discards two vessels, leaving ten final assays. All campaigns have scientific account (K1), sealed prediction (Q), and retrospective review (K2) stages; 238 meet the full source-and-assessment specification.

Retained table 2: A.2 Prediction targets and system-specific questions

System	Prediction responses	Reference target	Interval level
EC	Six conversion/efficiency responses, including score	Seeded final observation	80%
PA	Organic and aqueous product fractions	Noiseless pre-sampling fractions	90%
RX	Yield, conversion, selectivity, byproduct signal, risk, score	Five-observation mean; coverage over observations	80%
EQ / E, P and S	Normalized pH, dissociation, precipitation signal	Five-observation mean; coverage over observations	80%
C	Net recovery, purity, size index, fines fraction	Noiseless pre-assay responses	80%
P	Purity, original-charge recovery	Seeded final observation	80%

Table A2. Every campaign predicts twelve conditions. PA fractions are complementary, and its two decision questions reuse the conditions. Reference repeats reduce observation noise but do not add agent sessions.

Tables 3-18 report MAE / empirical interval coverage (%), averaged over five world sessions. All scheduled sessions remain included. PA intervals are nominally 90%; all others are 80%. Response scales and reference targets differ across systems.

Retained table 3: EC, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2983 / 55.0	0.2740 / 65.0	0.1272 / 88.3
Energy efficiency	0.3098 / 33.3	0.2595 / 51.7	0.1582 / 63.3
Faradaic efficiency	0.2580 / 46.7	0.2259 / 60.0	0.1356 / 73.3
Public score	0.1953 / 51.7	0.1939 / 60.0	0.1321 / 71.7
Selective product yield	0.0731 / 60.0	0.0817 / 38.3	0.0598 / 56.7
Transport efficiency	0.2602 / 45.0	0.2184 / 66.7	0.1328 / 76.7

Retained table 4: EC, E prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2908 / 63.3	0.2378 / 66.7	0.2661 / 73.3
Energy efficiency	0.2794 / 43.3	0.2541 / 31.7	0.2440 / 55.0
Faradaic efficiency	0.2506 / 53.3	0.2532 / 31.7	0.2315 / 56.7
Public score	0.1947 / 55.0	0.1561 / 58.3	0.1835 / 53.3
Selective product yield	0.0758 / 43.3	0.0623 / 53.3	0.0762 / 50.0
Transport efficiency	0.2478 / 58.3	0.2053 / 51.7	0.2217 / 60.0

Retained table 5: PA, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3
Organic fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3

Retained table 6: EC, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1902 / 68.3	0.1268 / 80.0	0.0428 / 93.3
Energy efficiency	0.1581 / 61.7	0.1651 / 68.3	0.0945 / 80.0
Faradaic efficiency	0.1458 / 65.0	0.1505 / 66.7	0.0877 / 78.3
Public score	0.1201 / 63.3	0.1452 / 73.3	0.0713 / 78.3
Selective product yield	0.0726 / 53.3	0.0573 / 70.0	0.0374 / 75.0
Transport efficiency	0.1495 / 65.0	0.1501 / 66.7	0.0846 / 78.3

Retained table 7: EC, E prior: optimization, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1403 / 85.0	0.1252 / 85.0	0.0864 / 90.0
Energy efficiency	0.1977 / 48.3	0.1307 / 60.0	0.1779 / 65.0
Faradaic efficiency	0.1502 / 76.7	0.1098 / 78.3	0.1515 / 71.7
Public score	0.1113 / 75.0	0.0996 / 71.7	0.1136 / 75.0
Selective product yield	0.0515 / 75.0	0.0498 / 56.7	0.0587 / 58.3
Transport efficiency	0.1420 / 80.0	0.1257 / 73.3	0.1570 / 68.3

Retained table 8: PA, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0251 / 100.0	0.0296 / 96.7	0.0207 / 100.0
Organic fraction	0.0211 / 100.0	0.0296 / 96.7	0.0207 / 100.0

Retained table 9: RX, P prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.0800 / 88.7	0.0791 / 73.3	0.0710 / 81.0
Conversion	0.0426 / 72.7	0.0321 / 70.0	0.0306 / 73.3
Safety risk	0.1031 / 58.3	0.0282 / 85.0	0.0370 / 81.7
Public score	0.1195 / 48.7	0.0501 / 83.3	0.0476 / 90.3
Selectivity	0.1198 / 58.0	0.0818 / 69.3	0.0714 / 77.3
Yield	0.1461 / 35.7	0.0708 / 76.7	0.0698 / 78.3

Retained table 10: RX, P prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1509 / 57.7	0.0805 / 73.0	0.1271 / 60.3
Conversion	0.0474 / 85.3	0.0319 / 82.7	0.1960 / 53.0
Safety risk	0.0984 / 56.7	0.1106 / 56.7	0.0811 / 68.3
Public score	0.1341 / 30.0	0.0818 / 73.0	0.0879 / 68.3
Selectivity	0.1900 / 31.0	0.0923 / 69.7	0.1360 / 64.7
Yield	0.2128 / 19.3	0.1060 / 65.7	0.1499 / 68.7

Retained table 11: RX, S prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1219 / 78.7	0.1212 / 75.3	0.1279 / 64.7
Conversion	0.0217 / 97.7	0.0266 / 95.3	0.0200 / 98.0
Safety risk	0.0664 / 80.0	0.0717 / 66.7	0.1276 / 43.3
Public score	0.0504 / 89.7	0.0816 / 61.3	0.0665 / 78.0
Selectivity	0.1280 / 72.7	0.1319 / 70.7	0.1400 / 55.0
Yield	0.1157 / 81.7	0.1196 / 75.7	0.1275 / 61.0

Retained table 12: RX, S prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1376 / 63.7	0.1490 / 67.7	0.1188 / 66.3
Conversion	0.0370 / 88.7	0.0505 / 82.7	0.0425 / 88.3
Safety risk	0.1000 / 45.0	0.1255 / 68.3	0.1133 / 45.0
Public score	0.0932 / 56.0	0.0917 / 58.0	0.0637 / 81.0
Selectivity	0.1574 / 56.7	0.1570 / 60.3	0.1175 / 62.3
Yield	0.1485 / 63.3	0.1422 / 66.3	0.0975 / 73.0

Retained table 13: EQ, P prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0129 / 96.3	0.0759 / 68.7	0.0646 / 74.7
Normalized pH	0.0031 / 94.3	0.0160 / 65.0	0.0150 / 68.7
Precipitation signal	0.0205 / 83.3	0.0376 / 73.3	0.0383 / 68.7

Retained table 14: EQ, S prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0292 / 91.7	0.0235 / 76.3	0.0122 / 87.7
Normalized pH	0.0097 / 87.3	0.0200 / 75.7	0.0203 / 77.7
Precipitation signal	0.0033 / 95.3	0.0042 / 95.7	0.0078 / 87.3

Retained table 15: EQ, E prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0128 / 83.7	0.0107 / 85.0	0.0134 / 79.0
Normalized pH	0.0233 / 80.7	0.0278 / 79.0	0.0202 / 82.0
Precipitation signal	0.0084 / 89.3	0.0116 / 92.0	0.0103 / 88.3

Retained table 16: C, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.3279 / 25.0	0.3096 / 38.3	0.2925 / 43.3
Crystal purity	0.0095 / 98.3	0.0497 / 60.0	0.0917 / 23.3
Particle-size index	0.0777 / 55.0	0.0624 / 58.3	0.0725 / 58.3
Net crystal recovery	0.1632 / 48.3	0.0606 / 83.3	0.0892 / 71.7

Retained table 17: C, E prior: delivery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.2281 / 51.7	0.2774 / 38.3	0.3046 / 41.7
Crystal purity	0.0197 / 95.0	0.0398 / 73.3	0.0508 / 55.0
Particle-size index	0.0566 / 71.7	0.0440 / 70.0	0.1023 / 48.3
Net crystal recovery	0.0911 / 61.7	0.0684 / 80.0	0.0786 / 73.3

Retained table 18: P, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Purity	0.2240 / 51.7	0.1883 / 53.3	0.2508 / 50.0
Recovery	0.0668 / 56.7	0.0687 / 51.7	0.0727 / 53.3

Retained table 19: C.1 Equilibrium: a reversal across public concentration regimes

Query group / arm	Macro MAE	Coverage (%)	Interval width	Source-mean MAE
Other nine / Opaque	0.01008	90.7	0.04706	0.00553
Other nine / Aligned	0.00465	91.3	0.03340	0.00518
Other nine / MisIndexed	0.00439	89.6	0.02798	0.00494
Three most dilute / Opaque	0.01833	93.3	0.15676	0.16961
Three most dilute / Aligned	0.15862	2.2	0.08084	0.16833
Three most dilute / MisIndexed	0.14397	13.8	0.09633	0.16836

Table C1. EQ/P subgroup results. Each row averages five worlds; rows reuse those worlds. The source-mean predictor is an additional post hoc diagnostic and obtains no new experiments. Its poor dilute-query performance shows why retaining the plateau is insufficient; its usefulness on the other nine depends on their response distribution.

Retained table 20: C.2 Crystallization: joint response changes under the same treatment

Budget / arm	Recovery MAE	Purity MAE	Recovery improves and purity worsens vs Opaque
12 / Opaque	0.16322	0.00952	Reference
12 / Aligned	0.06055	0.04969	5 / 5 worlds
12 / MisIndexed	0.08923	0.09166	5 / 5 worlds
24 / Opaque	0.09107	0.01975	Reference
24 / Aligned	0.06839	0.03983	2 / 5 worlds
24 / MisIndexed	0.07863	0.05080	3 / 5 worlds

Table C2. Joint directional comparisons use both response errors from the same world, arm and budget. At twelve batches, each joint direction also holds in all four pairs whose two campaigns conform to the source-assay contract. The direction weakens at 24 batches. A treatment can therefore be helpful on one response and harmful on another without implying a budget-invariant tradeoff.

Retained table 21: C.4 Original equilibrium campaigns: predictions and research chronology

World / arm	Public Q explanation	Dissociation [80% interval]	Reference
1 / Opaque	Piecewise weak acid	0.6990 [0.250, 0.900]	0.7160
1 / Aligned	Plateau / weak trend	0.1040 [0.044, 0.225]	0.7160
1 / MisIndexed	Plateau / weak trend	0.1092 [0.0772, 0.1412]	0.7160
2 / Opaque	Piecewise weak acid	0.6290 [0.450, 0.800]	0.6546
2 / Aligned	Plateau / weak trend	0.0804 [0.053, 0.108]	0.6546
2 / MisIndexed	Plateau / weak trend	0.0806 [0.0556, 0.1056]	0.6546
3 / Opaque	Piecewise weak acid	0.5630 [0.430, 0.700]	0.5797
3 / Aligned	Plateau / weak trend	0.1100 [0.045, 0.190]	0.5797
3 / MisIndexed	Mixed weak acid / plateau	0.4600 [0.100, 0.750]	0.5797
4 / Opaque	Qualitative departure	0.5200 [0.150, 0.880]	0.5173
4 / Aligned	Plateau / weak trend	0.0724 [0.040, 0.105]	0.5173
4 / MisIndexed	Plateau / weak trend	0.0962 [0.040, 0.450]	0.5173
5 / Opaque	Piecewise weak acid	0.4130 [0.200, 0.700]	0.4480
5 / Aligned	Plateau / weak trend	0.0790 [0.035, 0.130]	0.4480
5 / MisIndexed	Plateau / weak trend	0.0681 [0.040, 0.096]	0.4480

Table C3. Public extrapolation accounts and original same-context predictions. Categories are a single-author retrospective reading of public explanations, not internal reasoning traces or independent causal labels. Four Opaque accounts use

piecewise weak-acid reasoning; the fifth permits a qualitative departure. All five Aligned accounts extend a plateau or weak trend. The third-world MisIndexed mixed account is a counterexample to treating all dossier conditions alike.

Retained table 22: E.3 Complete crystallization baseline and interval comparisons

Table E1. Crystallization prediction errors and baseline comparisons.

Response	Agent MAE	Mean MAE	Nearest MAE	Wins: mean	Wins: nearest
Recovery	0.09185	0.17135	0.17308	26/30	26/30
Purity	0.04354	0.00482	0.00905	0/30	4/30
Size index	0.06923	0.02345	0.02708	4/30	7/30
Fines fraction	0.29000	0.31651	0.32426	21/30	20/30

MAE is averaged equally over all thirty campaigns in each response's original scale; errors are not ranked across responses. Recovery, purity and fines are fractions, while size is the declared size index. Mean and Nearest denote the public observation-mean and nearest-neighbour predictors. Wins count strictly lower agent MAE; neither comparison has ties. The source-assay shortfall remains included, and the five world instances are reused across arms and independent budget sessions.

Retained table 23: E.3 Complete crystallization baseline and interval comparisons

Table E2. Purity interval coverage and width by information arm and budget.

Budget	Information arm	Campaigns	Coverage (%)	Mean width (pp)
12	Opaque	5	98.3	6.81
12	Aligned	5	60.0	12.94
12	MisIndexed	5	23.3	15.53
24	Opaque	5	95.0	8.59
24	Aligned	5	73.3	12.37
24	MisIndexed	5	55.0	12.31

The original intervals have nominal 80% coverage and concern the pre-final-assay response target. Coverage and interval width are averaged over each campaign's twelve queries and then over its five-world arm-budget group. Each row contains sixty query targets but only five world clusters; queries are not independent world replicates. Width is expressed in percentage points of purity. All original intervals and source failures remain unchanged.

Retained table 24: E.4 Complete research-objective comparison

Table E3. All joint goal outcomes.

Higher retest	Lower score MAE	EC pairs	RX pairs
Optimization	Optimization	13	4
Optimization	Discovery	13	5
Discovery	Optimization	1	4
Discovery	Discovery	3	17

Both endpoints favour the same campaign in 16/30 EC and 21/30 RX pairs, and opposite campaigns in 14/30 and 9/30. Strict signs define these descriptive counts; near-zero differences are not thereby significant. Each system reuses five worlds. RX score MAE favours optimization in 8/30 pairs; its separate six-response macro MAE does so in 6/30.

Retained table 25: E.5 Strata and recovery sensitivity

Table E4. Endpoint orderings within design strata.

System	Factor	Level	Pairs	Same	Opposite
EC	World	W01	6	3	3
EC	World	W02	6	4	2
EC	World	W03	6	3	3
EC	World	W04	6	5	1
EC	World	W05	6	1	5
EC	Budget	12	15	7	8
EC	Budget	24	15	9	6
EC	Arm	Aligned	10	6	4
EC	Arm	MisIndexed	10	5	5
EC	Arm	Opaque	10	5	5
EC	Locus	E	30	16	14
RX	World	W01	6	4	2
RX	World	W02	6	4	2
RX	World	W03	6	3	3
RX	World	W04	6	4	2
RX	World	W05	6	6	0
RX	Budget	12	30	21	9
RX	Arm	Aligned	10	6	4
RX	Arm	MisIndexed	10	9	1
RX	Arm	Opaque	10	6	4
RX	Locus	P	15	12	3
RX	Locus	S	15	9	6

Same means that the higher-retest campaign also has lower MAE; opposite means it has higher MAE. These overlapping strata do not add independent tests. Budget rows compare goals at a fixed budget, not a twelve-to-twenty-four-batch contrast. E denotes entity priors, P parameter priors and S structural priors.

Retained table 26: E.6 All batches of the selected electrochemical pair

Table E5. Discovery source batches.

Batch	Pair	V	mA	Electrolysis / s	Score
1	S0/E0	-1.5	100	1200	0.000000
2	S1/E0	-1.5	100	1200	0.167530
3	S2/E0	-1.5	100	1200	0.000000
4	S3/E0	-1.5	100	1200	0.050903
5	S0/E0	0.5	100	1200	0.509008
6	S0/E0	1.5	100	1200	0.044293
7	S0/E0	2.5	100	1200	0.030644
8	S0/E1	0.5	100	1200	0.000000
9	S0/E2	0.5	100	1200	0.482517
10	S0/E3	0.5	100	1200	0.360679
11	S0/E2	0.5	100	1200+13200	0.606942
12	S0/E2	0.5	500	1200+13200	0.710658

Retained table 27: E.6 All batches of the selected electrochemical pair

Table E6. Optimization source batches.

Batch	Pair	V	mA	Electrolysis / s	Score
1	S0/E0	1	100	3600	0.535704
2	S1/E1	-1.5	300	7200	0.282124
3	S2/E2	1.5	300	7200	0.674932
4	S3/E3	1.5	300	7200	0.313717
5	S2/E0	1.5	300	7200	0.033808
6	S0/E2	1.5	300	7200	0.042443
7	S2/E2	1	300	7200	0.570485
8	S2/E2	2	300	7200	0.000000
9	S2/E2	1.5	500	7200	0.722737
10	S2/E2	1.5	500	10000	0.731956
11	S2/E2	1.4	500	9000	0.714786
12	S2/E2	1.5	400	12000	0.749194

Both campaigns select batch 12. Independent retests are 0.717587 for discovery and 0.756122 for optimization; twelve-query score MAEs are 0.088394 and 0.229397.

Retained table 28: E.8 Complete optimization forecasts

Table E7. Query settings, sealed forecasts and reference scores.

Q	Pair	V	mA	s	Forecast	80% interval	Reference
01	S0/E0	-0.8	100	600	0.050	0.000-0.180	0.032659
02	S0/E0	0.8	100	600	0.230	0.040-0.480	0.349491
03	S0/E0	-0.8	100	7200	0.250	0.080-0.440	0.358985
04	S0/E0	0.8	100	7200	0.570	0.430-0.680	0.572012
05	S0/E1	-0.8	100	7200	0.080	0.000-0.250	0.362359
06	S0/E1	0.8	100	7200	0.100	0.000-0.290	0.627599
07	S0/E3	-0.8	100	7200	0.070	0.000-0.230	0.347853
08	S0/E3	0.8	100	7200	0.090	0.000-0.280	0.542955
09	S2/E0	-0.8	100	7200	0.060	0.000-0.210	0.288226
10	S2/E0	0.8	100	7200	0.070	0.000-0.240	0.494376
11	S0/E0	-0.8	500	7200	0.170	0.040-0.360	0.276451
12	S0/E0	0.8	500	7200	0.550	0.390-0.680	0.344881

Score MAE is 0.229397; interval coverage is 5/12. The accurate Q04 prediction is retained alongside the large underestimates at Q06, Q08 and Q10. Source-to-query changes involve amount, volume and electrical settings, so these contrasts do not isolate one changed variable. Twelve query outcomes do not establish population calibration. No intermediate sealed prediction checkpoints were collected, so final prediction errors cannot be plotted as a within-session learning curve.

Retained table 29: F.1 Original intervals and campaign-specific observations

Table F1. Dissociation predictions at the most dilute recipe, with all values in percent.

World / arm	Source range	Point prediction	80% interval	Reference mean
1 / Opaque	6.18-8.34	69.90	[25.00, 90.00]	71.60
1 / Aligned	6.20-9.23	10.40	[4.40, 22.50]	71.60
1 / MisIndexed	6.11-9.25	10.92	[7.72, 14.12]	71.60
2 / Opaque	6.17-8.57	62.90	[45.00, 80.00]	65.46
2 / Aligned	6.12-8.06	8.04	[5.30, 10.80]	65.46
2 / MisIndexed	6.23-7.98	8.06	[5.56, 10.56]	65.46
3 / Opaque	5.94-8.22	56.30	[43.00, 70.00]	57.97
3 / Aligned	6.34-8.02	11.00	[4.50, 19.00]	57.97

World / arm	Source range	Point prediction	80% interval	Reference mean
3 / MisIndexed	6.16-7.94	46.00	[10.00, 75.00]	57.97
4 / Opaque	6.19-7.91	52.00	[15.00, 88.00]	51.73
4 / Aligned	5.82-8.14	7.24	[4.00, 10.50]	51.73
4 / MisIndexed	5.81-8.22	9.62	[4.00, 45.00]	51.73
5 / Opaque	5.49-7.48	41.30	[20.00, 70.00]	44.80
5 / Aligned	6.44-7.99	7.90	[3.50, 13.00]	44.80
5 / MisIndexed	6.58-7.79	6.81	[4.00, 9.60]	44.80

The recipe contains 1 micromole in 75 mL (nominal concentration 13.3 μM). The displayed source ranges are specific to each campaign. World 3 MisIndexed retains its mixed-explanation departure from the plateau. Public explanation categories appear in retained table 21 above.

Retained table 30: F.2 Sol interval and regime comparisons

Supplementary Table F2. GPT-5.6 Sol equilibrium predictions across test regions.

Arm	Remaining nine: MAE	Lowest three: MAE	Remaining nine: coverage	Lowest three: coverage
Opaque	0.01008	0.01833	90.7%	93.3%
Aligned	0.00465	0.15862	91.3%	2.2%
MisIndexed	0.00439	0.14397	89.6%	13.8%

Values average five worlds and three response targets per group: normalized pH, dissociation fraction and precipitation signal. Coverage uses original 80% prediction intervals, with 675 reference-containment judgments per arm for the remaining nine conditions and 225 for the three lowest-concentration conditions; these judgments are not independent world replicates. The concentration grouping is post hoc and exploratory; the remaining nine conditions also include boundaries.

Retained table 31: F.3 Complete five-model comparison

Supplementary Table F3. Equilibrium predictions from five medium model configurations.

Model	Arm	Remaining nine: MAE	Lowest three: MAE	Lowest three: coverage
GPT-5.5	Opaque	0.00647	0.16325	6.7%
GPT-5.5	Aligned	0.00638	0.16146	2.7%
GPT-5.5	MisIndexed	0.00461	0.13604	13.3%
GPT-5.6 Luna	Opaque	0.09867	0.15240	48.4%
GPT-5.6 Luna	Aligned	0.04997	0.13319	31.1%
GPT-5.6 Luna	MisIndexed	0.05494	0.12556	44.4%
GPT-5.6 Terra	Opaque	0.03268	0.14611	67.1%
GPT-5.6 Terra	Aligned	0.01545	0.11466	26.2%
GPT-5.6 Terra	MisIndexed	0.01369	0.12130	41.8%
GPT-5.6 Sol	Opaque	0.01008	0.01833	93.3%
GPT-5.6 Sol	Aligned	0.00465	0.15862	2.2%
GPT-5.6 Sol	MisIndexed	0.00439	0.14397	13.8%
GPT-6 Astra	Opaque	0.00244	0.04040	82.7%
GPT-6 Astra	Aligned	0.00384	0.03853	82.7%
GPT-6 Astra	MisIndexed	0.00234	0.01857	92.4%

Each row contains five independent source campaigns, equally weighting the five worlds; worlds recur across information conditions and models. Each campaign predicts pH/14, dissociation fraction and precipitation signal for twelve queries. The three lowest-concentration conditions are the same fixed query group; the remaining nine also include boundary and unexplored conditions. Coverage concerns original 80% intervals and five reference observations, not confidence intervals or independent model replications. All three arms are retained; lower MisIndexed error does not itself establish a causal benefit of incorrect knowledge.

Retained table 32: G.1 Equilibrium reversal by response

Table G1. Original prediction MAE by response and concentration group.

Response	Group	Opaque	Aligned	MisIndexed	Lower / higher
pH / 14	Other nine	0.00261	0.00269	0.00254	2 / 3 of 5
pH / 14	Dilute three	0.00448	0.05576	0.05230	0 / 5 of 5
Dissociation fraction	Other nine	0.00839	0.00709	0.00660	3 / 2 of 5
Dissociation fraction	Dilute three	0.02640	0.28229	0.23861	0 / 5 of 5
Precipitation signal	Other nine	0.01923	0.00417	0.00403	5 / 0 of 5
Precipitation signal	Dilute three	0.02411	0.13783	0.14099	0 / 5 of 5

The final column counts worlds with lower/higher Aligned MAE than Opaque. Signed response differences divided by three sum to the macro-error difference. They quantify the contribution on the declared normalized scales; they do not establish a shared failure mechanism or physical comparability of targets.

Retained table 33: G.2 Separating purity offset from variation error

Table G2. Purity offset and variation diagnostics.

For each campaign, prediction error decomposes as MSE $= \overline{p-y}^{\,2} + \frac{1}{12}\sum_i[(p_i-\bar{p})-(y_i-\bar{y})]^2$. Centred RMSE is the square root of the second term. Demeaning uses withheld references for diagnosis, not as an available predictor.

Batches	Arm	N	Bias	cRMSE	Ref. SD	Pred. SD	Offset share
Both	All	30	-4.289	1.889	0.598	1.968	87.6%
12	Opaque	5	-0.806	0.970	0.598	1.031	45.3%
12	Aligned	5	-4.920	1.837	0.598	1.791	89.8%
12	MisIndexed	5	-9.166	2.843	0.598	2.977	91.3%
24	Opaque	5	-1.816	1.682	0.598	1.649	57.4%
24	Aligned	5	-3.948	1.797	0.598	1.983	85.6%
24	MisIndexed	5	-5.080	2.206	0.598	2.376	85.3%

N is the campaign count; cRMSE denotes centred RMSE. Bias, cRMSE and both standard deviations are campaign means in percentage points. Offset share is the ratio of summed squared campaign biases to summed campaign MSEs, with equal query counts; it is not the mean campaign fraction. The median campaign fraction is 77.6%. Mean bias is negative in 30 of 30 campaigns. Centred RMSE exceeds reference SD in 29 of 30. These comparisons describe forecast outputs and do not identify an internal revision process. No new agent or simulator calls were made.

Retained table 34: H.2 A concrete three-arm information example

Table H1. Public prior content in one matched physical world.

Supplied field	Opaque	Aligned	MisIndexed
Effective pKa interval (80%)	Absent	[4.6094, 4.7094]	[5.3294, 5.4294]
Dilution change in pH/14	Absent	+0.003494987	+0.000875889
Dilution change in dissociation	Absent	+0.008187416	+0.001884676
Nominal confidence	Absent	0.8	0.8

The common dilution anchor is 0.001 mol acid at 298.15 K, with volume increasing from 0.018 to 0.054 L. Both supplied records identify the source as an “independent bounded archival fit” and qualify the claim as a “local effective relationship; not a universal aqueous-chemistry law”. The pKa intervals above are rounded for display; the input contains their full precision. Aligned therefore supplies correct partial local information, not the evaluator’s complete process equations, query answers or an instruction to enter the most dilute test region. The experiment contrasts the resulting complete research processes, including possible changes to evidence acquisition and interpretation.

Executed process relationships

The following equations describe the retained virtual-world laws, not mechanisms inferred by the participants. The initial-charge final-assay recovery formula below also describes the checked full-process observation route in the repeated cohort. The internal stage-recovery metric and mismatched participant description are distinguished above; they must not be substituted for the observed/scored response.

Equilibrium response model

The parameter-prior study uses a bounded monoprotic acid-base model followed by a precipitation calculation. These are the process relations used for the retained campaigns and references. They are distinct from explanations proposed by the agents. In particular, an agent's proposed active-pool cap is not the implemented source of the dissociation plateau.

Let n be the current ledger amount of the initial reactant, V the current sample volume and $C = n/V$. The background-ion amounts are $n_+ = \min(0.020, 0.15n)$ and $n_- = \min(0.020, 0.08n)$, in moles. With $H = [\mathrm{H^+}]$, $K_a = 10^{-\mathrm{p}K_a}$ and activity-ratio factor g, the solver obtains

$$[\mathrm{A^-}] = \frac{CK_a}{K_a + gH}, \qquad [\mathrm{OH^-}] = \frac{K_w(T)}{H},$$

subject to charge balance,

$$H + \frac{n_+}{V} - \frac{K_w(T)}{H} - [\mathrm{A^-}] - \frac{n_-}{V} = 0.$$

The reported responses are $\mathrm{pH}/14 = -\log_{10}(H)/14$ and dissociation fraction $\alpha = [\mathrm{A^-}]/C$, clipped to the public unit interval. In the uncapped background-ion range, charge balance implies $\alpha = 0.07 + H/C - K_w/(HC)$. At sufficiently high concentration the last two terms are small, giving a local plateau near 7%; dilution makes their contribution appreciable. The plateau and departure therefore arise under the same fixed equations. The five effective pKa values span 4.6594 to 5.3794 in increments of 0.18; the default activity-ratio factor is one.

Precipitation is calculated sequentially from effective ion inventories $a = n_+ + 0.10n_{\mathrm{A^-}}$ and $b = n_- + 0.08n_{\mathrm{HA}}$. When supersaturated, a 1:1 removal x, bounded by the limiting inventory, satisfies $(a - x)(b - x)/V^2 = K_{sp}$, with default $K_{sp} = 1.8 \times 10^{-10}$. The public signal is x/n, clipped to $[0, 1]$. This hook does not feed precipitation back into the acid-base solution. Thus the three public responses are related but should not be interpreted as three independent measurements of one failure mechanism.

Nominal concentration in the retained source comparisons is a recipe descriptor: total nominal reagent additions divided by total solvent additions. The solver instead uses the ledger state at measurement, after any sample consumption and other operations. The scatterplot combines distinct recipes and histories; it is not a controlled one-dimensional titration curve. The three most dilute test recipes have nominal concentrations of approximately 1.33×10^{-4}, 1.33×10^{-5} and 1.67×10^{-4} M. No source assay in these fifteen Sol campaigns reaches those concentrations.

Crystallization response model

The retained crystallization block composes upstream reaction, thermal operations, seeding, crystal nucleation and growth, filtration and measurement. It uses the latent-material process family and its linear supersaturation-dependent impurity transfer. Solid product, solid impurity and the particle population persist across operations; dissolved and solid inventories are accounted for separately.

The temperature-dependent solubility follows

$$c^*(T) = c^*_{\mathrm{ref}} \exp\left[-\frac{\Delta H}{R}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)\right],$$

with $T_{\mathrm{ref}} = 298.15\,\mathrm{K}$, $\Delta H = 20{,}000\,\mathrm{J\,mol^{-1}}$ and a declared temperature domain of 250-430 K. World and solvent properties scale the reference solubility and kinetics. For relative supersaturation $s = \max(C/c^* - 1, 0)$, primary nucleation scales as $B = k_b s^2$ and linear growth as $G = k_g s$. Before world and solvent multipliers, the coefficients are $2 \times 10^7\ \mathrm{L^{-1}\,s^{-1}}$ for nucleation and $2 \times 10^{-8}\ \mathrm{m\,s^{-1}}$ for growth. Nucleation and growth consume available dissolved target and update the population distribution; they are not independent score adjustments.

Each crystallized target increment transfers impurity according to

$$\Delta n_{I,s} = \min(n_{I,l},\ \Delta n_{P,s}\, k_{\mathrm{occ}}(1 + 0.5s)),$$

where $k_{\mathrm{occ}} = 0.02$ times the solvent's occlusion multiplier. The four nominal solvent multipliers are 1.20, 0.80, 0.65 and 1.05; world-fixed material residuals further modulate these properties. Filtration retains 96% of target solids and 92% of impurity solids. The final-assay purity is the solid-target mole fraction $n_{P,s}/(n_{P,s} + n_{I,s})$. Seed contributes to target purity but is subtracted in the full-process recovery numerator; final-assay recovery is normalized by the initial reagent charge.

These relations allow thermal and seeding histories to alter the amount and size distribution of crystals while impurity incorporation remains modest. They explain how recovery and particle responses can vary without requiring a comparable purity change. The approximately 98.5% purity level is an empirical property of the retained source and query conditions, not a hard-coded value or a universal invariant of the model. The present analysis does not isolate individual process contributions to that stability. The retained diagnostics test whether the forecasts preserve the level and variation of the retained purity references.